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SYSTEMS AND METHODS FOR DOUBLE PISTON EFFECT BALL VALVES

Abstract

A ball valve may include a body having an upstream end and a downstream end. A ball may be within the body between the two ends, the ball being rotatable about a vertical axis to move between an open position and a closed position. A ball valve may include a stem with a body and an actuator portion that enables the ball to be rotated between the open and closed positions. A ball valve may include a seat assembly with a first end facing towards the ball and a second end facing away from the ball, the seat assembly including a seat, the seat being in contact with the ball; a lantern ring, the lantern ring positioned closer to the second end of the seat assembly than the first end of the seat assembly; and a seal, the seal positioned between the seat and the lantern ring.

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Background/Summary

TECHNICAL FIELD

[0001] Aspects of the present disclosure relate generally to valves and valve systems. In particular, aspects of the present disclosure relate to methods and systems for ball valve assemblies, such as ball valve assemblies that include single piston effect and/or double piston effect seats.

BACKGROUND

[0002] Valve devices, such as ball valve assemblies, are useful for fluid handling in, for example, the oil and gas, power, chemical, water works, waste water, and manufacturing industries. Ball valves assemblies are useful to selectively permit or block the flow of large volumes of fluid. Ball valves are robust systems that operate in harsh environments and under severe conditions. In order to perform in under these circumstances, valve systems include components that are resistant to debris, corrosion, and wear. Additionally, components are regularly inspected and maintained to ensure continued operation of the valve. Depending on the application, a ball valve assembly may include a single piston effect and/or double piston effect seat. The valve body is specifically designed for the type of piston effect the valve will have. Single piston effect seats act differently under pressure than double piston effect seats, allowing for pressure to vent. In double piston effect applications, the movement of seals due to the high pressure can cause damage to the seals, which may cause improper sealing in the valve.

SUMMARY

[0003] Additional objects and advantages of the disclosed embodiments will be set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practice of the disclosed embodiments. The objects and advantages of the disclosed embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims.

[0004] In some aspects, the techniques described herein relate to a ball valve system, including: a body having an upstream end and a downstream end; a ball within the body between the upstream end and the downstream end, the ball being rotatable about a vertical axis to move between an open position and a closed position; a stem with a body and an actuator portion, and wherein the stem is coupled to the ball such that the actuator portion of the stem is configured to rotate the ball between the open position and the closed position; and a seat assembly with a first end facing towards the ball and a second end facing away from the ball, the seat assembly including: a seat, the seat being in contact with the ball; a lantern ring, the lantern ring positioned closer to the second end of the seat assembly than the first end of the seat assembly; and a seal, the seal positioned between the seat and the lantern ring.

[0005] In some aspects, the techniques described herein relate to a ball valve system, wherein the actuator portion of the stem includes a handwheel.

[0006] In some aspects, the techniques described herein relate to a ball valve system, wherein an opening at an inner radial surface of the seat assembly is in fluid communication with a central portion of the lantern ring.

[0007] In some aspects, the techniques described herein relate to a ball valve system, wherein the lantern ring is secured in a slot formed between the body and the seat.

[0008] In some aspects, the techniques described herein relate to a ball valve system, wherein the slot is formed as a recess at an outer radial surface of the seat assembly.

[0009] In some aspects, the techniques described herein relate to a ball valve system, wherein the recess extends from a first axial end of the seat toward a second axial end of the seat.

[0010] In some aspects, the techniques described herein relate to a ball valve system, wherein the seat, the lantern ring, and the seal are positioned in such a way that the seat assembly creates a

double piston effect.

[0011] In some aspects, the techniques described herein relate to a ball valve system, wherein the lantern ring limits movement of the seal such that the seal is prevented from movement to the second end of the seat assembly.

[0012] In some aspects, the techniques described herein relate to a method of assembling a ball valve system, the method including: securing a ball within a body, the body including an upstream end and a downstream end; inserting a stem within the ball, the stem including an actuator portion; rotating the ball, via the actuator portion of the stem, between an open position and a closed position; securing a seat assembly with a first end facing towards the ball and a second end facing away from the ball, the seat assembly including: a seat, the seat being in contact with the ball; a lantern ring, the lantern ring positioned closer to the second end of the seat assembly than the first end of the seat assembly; and a seal, the seal positioned between the seat and the lantern ring.

[0013] In some aspects, the techniques described herein relate to a method, wherein the actuator portion of the stem includes a handwheel.

[0014] In some aspects, the techniques described herein relate to a method, wherein a radial opening of the seat assembly is in fluid communication with the center of the lantern ring.

[0015] In some aspects, the techniques described herein relate to a method, wherein the lantern ring is secured in a slot formed between the body and the seat.

[0016] In some aspects, the techniques described herein relate to a method, wherein the slot is formed as a recess at an outer radial surface of the seat assembly.

[0017] In some aspects, the techniques described herein relate to a method, wherein the recess extends from an axial end of the seat.

[0018] In some aspects, the techniques described herein relate to a method, wherein the seat, the lantern ring, and the seal are positioned in such a way that the seat assembly creates a double piston effect.

[0019] In some aspects, the techniques described herein relate to a method, wherein the lantern ring limits movement of the seal such that the seal is prevented from movement to the second end of the seat assembly.

[0020] In some aspects, the techniques described herein relate to a ball valve system, including: a body having an upstream end and a downstream end; a ball within the body between the upstream end and the downstream end, the ball being rotatable about a vertical axis to move between an open position and a closed position; a stem with a body and an actuator portion, wherein the body of the stem is positioned substantially parallel to the vertical axis, and wherein the stem is coupled to the ball such that the actuator portion of the stem is configured to rotate the ball between the open position and the closed position; and a seat assembly with a first end facing towards the ball and a second end facing away from the ball, the seat assembly including: a seat, the seat being in contact with the ball; a lantern ring, the lantern ring positioned closer to the second end of the seat assembly than the first end of the seat assembly; and a seal, the seal positioned between the seat and the lantern ring, wherein the seat, lantern ring, and seal are positioned in such a way that the seat assembly creates a double piston effect, and wherein the lantern ring limits movement of the seal such that the seal is prevented from movement to the second end of the seat assembly.

[0021] In some aspects, the techniques described herein relate to a ball valve system, wherein the lantern ring is secured in a slot formed between the body and the seat.

[0022] In some aspects, the techniques described herein relate to a ball valve system, wherein the slot is formed as a recess at an outer radial surface of the seat assembly.

[0023] In some aspects, the techniques described herein relate to a ball valve system, wherein the recess extends from an axial end of the seat.

[0024] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

[0026] FIG. 1 is a perspective view of a ball valve system, according to aspects of the present disclosure;

[0027] FIG. 2 is a side view of the ball valve system of FIG. 1;

[0028] FIG. 3 is a cross-sectional view of the ball valve system of FIG. 1;

[0029] FIG. 4 is a detailed view of a portion of a double piston effect seat contacting a ball of a ball valve system, according to aspects of the present disclosure.

[0030] FIG. 5 is a detailed view of a portion of a single piston effect seat contacting a ball of a ball valve system, according to aspects of the present disclosure.

DETAILED DESCRIPTION

[0031] Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. Moreover, in this disclosure, relative terms, such as, for example, “about,” “substantially,” “generally,” and “approximately” are used to indicate a possible variation of $\pm 10\%$ in the stated value.

[0032] FIGS. 1 and 2 are perspective and side views, respectively, of a ball valve assembly or ball valve system **100**, according to aspects of the present disclosure. In some aspects, ball valve system **100** may be a double block and bleed (DBB) ball valve system. In other aspects, ball valve system **100** may be a double isolation and bleed type 1 (DIB-1) ball valve system. In yet other aspects, ball valve system **100** may be a double isolation and bleed type 2 (DIB-2) ball valve system.

[0033] Ball valve system **100** may include a body **102** extending from an upstream end **104** to a downstream end **106**. Annular flanges or ports **108**, formed at upstream and downstream ends **104** and **106**, may be sized and shaped for connection to upstream and downstream pipeline components, and may include a series of bolt holes to allow ports **108** to facilitate leak-free connections to these components.

[0034] A mounting plate **112** may be secured to a top surface of body **102**. A trunnion **114** may be secured to a bottom surface of body **102**. In some aspects, ball valve system **100** may also include a manual actuator in the form of gear operator **116** and handwheel **118**, operator **116** and/or handwheel **118** forming an actuator portion of a stem **300** (described below). The gear operator **116** may be secured to the ball valve system **100** by mounting plate **112**. Mounting plate **112** may be fixed to body **102** by a series of fasteners, such as threaded bolts or studs, and fixing members, such as nuts.

[0035] Body **102** may be formed of any suitable corrosion-resistant material, such as a metal material. In particular, body **102** may be formed by carbon steel, for example. Mounting plate **112**, like body **102**, may be formed by a metal material, such as carbon steel. A gland **120** included between body **102** and mounting plate **112** may facilitate leak-free operation of stem **300** (shown in FIG. 3). Gland **120** may include, for example, packing for preventing leaks, as well as retaining features for securing this packing adjacent to stem **300**.

[0036] As shown in FIG. 3, gland **120** and/or mounting plate **112** of ball valve system **100** may support and secure stem **300** therein. A portion of stem **300** above gland **120** and mounting plate **112** may be connected to gear operator **116** (and indirectly to handwheel **118**). One or more

embodiments of the present disclosure may include a pneumatic or hydraulic actuator instead of or in addition to handwheel **118**. Actuation of handwheel **118** may be configured to rotate ball **302** (shown in FIG. **3**) between an open and a closed position.

[0037] During operation, ball valve system **100** selectively permits flow from upstream end **104** to downstream end **106**. For example, ball valve system **100** may be in an open position where upstream end **104** is in fluid communication with downstream end **106** or ball valve system **100** may be in a closed position where upstream end **104** is not in fluid communication with downstream end **106**. When transitioning from an open position to a closed position, fluid or gas from upstream end **104** may tend to enter, and subsequently become trapped, inside of the section of ball valve system **100** that houses ball **302** (shown in FIG. **3** and described below) between upstream end **104** and downstream end **106**. Ball valve system **100** is capable of compensating for this unintentional trapped fluid or gas.

[0038] While a single piston effect seat allows for the pressure to escape into downstream end **106** via self-venting, a double piston effect seat may not be configured for self-venting. For example, a single piston effect seat may shift away from the ball at a threshold pressure difference while a double piston effect seat may contact the ball regardless of the pressure difference. As such, at least some examples of ball valve system **100** may benefit from manual venting of pressure build-up. A body relief valve **110** (e.g., a one-way valve with a pre-set opening force) may be provided so as to allow for venting of pressure inside of ball valve system **100** if the pressure inside of ball valve system **100** exceeds a desired level.

[0039] FIG. **3** is a cross-sectional view of ball valve system **100**. A flow path **314** extends through upstream end **104** and downstream end **106** within an inner surface **312** of the wall **308** of body **102**. In operation, a liquid may flow from upstream end **104** to downstream end **106** when the ball valve system **100** is open.

[0040] As shown in FIG. **3**, first end **304** of stem **300** may extend vertically above an outer surface **310** of wall **308**. A second end **306** of stem **300** may be positioned within body **102**. More specifically, second end **306** may interface with and contact ball **302** and extend into a cavity **316** of ball **302**. In this way, actuation of handwheel **118** rotates stem **300** which in turn rotates ball **302**. Stem **300** may be formed of metal, such as stainless steel, and, in particular, 17-4 stainless steel.

[0041] Ball **302** may sit on trunnion **114** or another type of bearing. Trunnion **114** may extend into a cavity **318** of ball **302**. Trunnion **114** may advantageously absorb pressure, assist with lowering operating torque for stem **300**, assist with keeping ball **302** firmly in place, and define an axis of rotation for ball **302**. Trunnion **114** may be formed of a low-friction material configured to allow substantially frictionless rotation of ball **302**. A support portion of trunnion **114** may be removable at a bottom of ball valve system **100** which may facilitate maintenance and/or replacement of trunnion **114**. The trunnion **114** support may either be formed integrally with trunnion **114**, as shown in FIG. **3**, or separately of trunnion **114**.

[0042] Ball **302** may be rotatable between an open position and a closed position to respectively permit or block a flow of fluid from upstream end **104** to downstream end **106**. Ball **302** may include a central channel extending through ball **302** such that, when ball **302** is in the open position, the channel extends approximately parallel to, or aligned with, flow path **314**, and when ball **302** is in the closed position, the channel extends approximately perpendicular to flow path **314**. When in the closed position, a portion of ball **302** entirely blocks the flow path **314**, thus the channel is not in fluid communication with flow path **314**.

[0043] FIG. **3** also shows a seat assembly **320** contacting ball **302** from the upstream end **104** side and a seat assembly **322** contacting ball **302** from the downstream end **106** side. In some embodiments, seat assembly **320** and seat assembly **322** may be substantially similar. In other embodiments, seat assembly **320** and seat assembly **322** may be different from one another. For example, seat assembly **320** and seat assembly **322** may each provide a single piston effect (e.g., when ball valve system **100** is a DBB ball valve system), seat assembly **320** and seat assembly **322**

may each provide a double piston effect (e.g., when ball valve system **100** is a DIB-1 ball valve system), or seat assembly **320** may provide a single piston effect while seat assembly **322** may provide a double piston effect (e.g., when ball valve system **100** is a DIB-2 ball valve system). [0044] FIG. **4** is a detailed view of a portion of a double piston effect seat contacting a ball of a ball valve system, according to aspects of the present disclosure. Seat assembly **400** may be substantially similar to seat assembly **320** and/or seat assembly **322** shown in FIG. **3**. As such, FIGS. **4** and **5** may be considered different embodiments of “Detail A” shown in FIG. **3**. In aspects of the present invention, seat assembly **400** is a ring shape so that seat assembly **400** does not block flow path **314** while an outer circumference of seat assembly **400** is in contact with inner surface **312**.

[0045] Seat assembly **400** may include a seat **402**, a seat insert **404**, one or more seals **406**, a lantern ring **408**, and a spring **410**. Seat insert **404** may be inserted into a cavity of seat **402** such that seat **402** substantially surrounds seat insert **404**. Seat insert **404** may then contact a portion of ball **302** when ball **302** is in a closed position (i.e., when ball **302** is not permitting fluid communication between upstream end **104** and downstream end **106**). In some embodiments, seat insert **404** is the only component of seat assembly **400** that is in contact with ball **302**. In other embodiments, seat insert **404** is one of the components of seat assembly **400** that is in contact with ball **302**. Seat assembly **400** may include a sealing surface (e.g., a chamfered surface) configured to contact and seal ball **302**. If desired, this surface may be included on a seal of the one or more seals **406**, or may be included on seat insert **404**.

[0046] Seat assembly **400** may also include one or more recesses for inserting seals, such as gaskets or O-rings, between seat assembly **400** and inner surface **312**. For example, a slot **418** with a radial width W may be provided for a seal **406**, as shown in FIG. **4**. Seal **406** is shown in an uncompressed state and may occupy the width W of the gap when compressed. One or more seals **406** may be positioned near the center of seat assembly **400** between a first side **414** and a second side **416**. Furthermore, seals **406** may be between seat insert **404** and lantern ring **408**. Seals **406**, such as an O-ring, may facilitate the formation of a seal between seat assembly **400** and inner surface **312**.

[0047] As shown, lantern ring **408** may have opposing enlarged ends and a narrowed central portion, in contrast with an o-ring having a circular cross-sectional shape. In some aspects, the narrowed central portion of lantern ring **408** may have a series of circumferentially spaced apart bores, providing fluid communication above on opposite sides of lantern ring **408** within slot **418**.

[0048] In the example illustrated in FIG. **4**, seal **406** is secured in the slot **418** formed by the space between seat **402** and wall **308**. Slot **418** may be formed as an annular recess on an outer radial surface of seat **402**. This recess may extend from an outer axial end of seat **402**, this outer axial end being an upstream end in the case of an upstream seat **402** and a downstream end in the case of a downstream seat **402**.

[0049] A sealant injection system may be used in ball valve system **100** to provide emergency sealant. For example, emergency or backup sealing may be used when the sealing rings (e.g., seat insert **404** or lantern ring **408**) become damaged or otherwise no longer seal properly. That is, the sealant provided by sealant injection system may be a secondary, temporary sealing solution for situations when the primary (e.g., the seat assembly) system fails to seal properly.

[0050] The sealant may axially travel through an axial travel path **420a** axially to lantern ring **408** to a radially-extending outlet **420b**. Outlet **420b** may be formed as a hole extending through an outer radial surface of seat **402** to axial travel path **420a**. Sealant exiting radially-extending outlet **420b** may contact an inner circumferential surface of lantern ring **408** and/or fill lantern ring **408** by travelling through radial bores (not shown) extending through a narrowed central portion of lantern ring **408**. In this way, ball valve system **100** is protected from failure.

[0051] Spring **410** may be in contact with a side surface **412** of seat **402** and inner surface **312**. When ball valve system **100** is in use and ball **302** is in a closed position, the pressure difference

between first side **414** of seat assembly **400** facing towards ball **302** and second side **416** of seat assembly **400** facing away from ball **302** may cause the spring to compress. However, because seat assembly **400** is a double piston effect seat, seat assembly **400** does not shift away from ball **302**. Instead, seat assembly **400** remains in contact with ball **302**, thereby continuing to separate the fluid or gas (and therefore the pressure) on either side. Components of seat assembly **400**, like seal **406**, may shift during the compression of spring **410**. Lantern ring **408** may help protect one or more seals **406** from damage while shifting.

[0052] Lantern ring **408** may be positioned close to second side **416** within slot **418**. More specifically, lantern ring **408** may be closer to second side **416** than one or more seals **406** so that, when seal **406** shifts towards second side **416** due to the pressure difference, lantern ring **408** will interact with one or more seals **406**. Specifically, lantern ring **408** may act as a seal spacer or cushion so a seal **406** does not slide all the way to second side **416** when pressure acts against it in the double piston effect condition. In other words, lantern ring **408** limits the movement of a seal **406** to protect a seal **406** from twisting and/or binding.

[0053] FIG. 5 is a detailed view of a portion of a single piston effect seat contacting a ball of a ball valve system, according to aspects of the present disclosure. Seat assembly **500** may be substantially similar to seat assembly **320** and/or seat assembly **322** shown in FIG. 3. FIG. 5 may be considered different an embodiment of “Detail A” shown in FIG. 3. In aspects of the present invention, seat assembly **500** is a ring shape so that seat assembly **500** does not block flow path **314** while an outer circumference of seat assembly **500** is in contact with inner surface **312**.

[0054] Seat assembly **500** may include a seat **502**, a seat insert **504**, one or more seals **506**, and a spring **508**. Seat insert **504** may be inserted into a cavity of seat **502** such that seat **502** substantially surrounds seat insert **504**. Seat insert **504** may contact a portion of ball **302** when ball **302** is in a closed position (i.e., when ball **302** is not permitting fluid communication between upstream end **104** and downstream end **106**). In some embodiments, seat insert **504** is the only component of seat assembly **500** that is in contact with ball **302**. In other embodiments, ball **302** is one of the components of seat assembly **500** that is in contact with ball **302**. Seat assembly **500** may include a sealing surface (e.g., a chamfered surface) configured to contact and seal ball **302**. If desired, this surface may be included on a seal **506**, or may be included on seat insert **504**. Seat assembly **500** may also include one or more recesses for inserting seals, such as gaskets or O-rings, between seat assembly **500** and inner surface **312**. One or more seals **506** may be positioned near the center of seat assembly **500** between a first side **512** and a second side **514**. One or more seals **506**, such as an O-ring, may facilitate the formation of a seal between seat assembly **500** and inner surface **312**.

[0055] Spring **508** may be in contact with a side surface **510** of seat **502** and inner surface **312**. When ball valve system **100** is in use and ball **302** is in a closed position, the pressure difference between first side **512** of seat assembly **500** facing towards ball **302** and second side **514** of seat assembly **500** facing away from ball **302** may cause the spring **508** to compress. Spring **508** may therefore cause seat assembly **500** to shift towards the direction of second side **514** when there is enough pressure built up on first side **512**. The shift of seat assembly **500** may cause seat assembly **500** to no longer contact ball **302**, thus allowing for the pressure from first side **512** to vent to second side **514**.

[0056] Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

Claims

1. A ball valve system, comprising: a body having an upstream end and a downstream end; a ball within the body between the upstream end and the downstream end, the ball being rotatable about a

vertical axis to move between an open position and a closed position; a stem with a body and an actuator portion, and wherein the stem is coupled to the ball such that the actuator portion of the stem is configured to rotate the ball between the open position and the closed position; and a seat assembly with a first end facing towards the ball and a second end facing away from the ball, the seat assembly including: a seat, the seat being in contact with the ball; a lantern ring, the lantern ring positioned closer to the second end of the seat assembly than the first end of the seat assembly; and a seal, the seal positioned between the seat and the lantern ring.

2. The ball valve system of claim 1, wherein the actuator portion of the stem comprises a handwheel.

3. The ball valve system of claim 1, wherein an opening at an inner radial surface of the seat assembly is in fluid communication with a central portion of the lantern ring.

4. The ball valve system of claim 1, wherein the lantern ring is secured in a slot formed between the body and the seat.

5. The ball valve system of claim 4, wherein the slot is formed as a recess at an outer radial surface of the seat assembly.

6. The ball valve system of claim 5, wherein the recess extends from a first axial end of the seat toward a second axial end of the seat.

7. The ball valve system of claim 1, wherein the seat, the lantern ring, and the seal are positioned in such a way that the seat assembly creates a double piston effect.

8. The ball valve system of claim 1, wherein the lantern ring limits movement of the seal such that the seal is prevented from movement to the second end of the seat assembly.

9. A method of assembling a ball valve system, the method comprising: securing a ball within a body, the body including an upstream end and a downstream end; inserting a stem within the ball, the stem including an actuator portion; rotating the ball, via the actuator portion of the stem, between an open position and a closed position; securing a seat assembly with a first end facing towards the ball and a second end facing away from the ball, the seat assembly including: a seat, the seat being in contact with the ball; a lantern ring, the lantern ring positioned closer to the second end of the seat assembly than the first end of the seat assembly; and a seal, the seal positioned between the seat and the lantern ring.

10. The method of claim 9, wherein the actuator portion of the stem comprises a handwheel.

11. The method of claim 9, wherein a radial opening of the seat assembly is in fluid communication with the center of the lantern ring.

12. The method of claim 9, wherein the lantern ring is secured in a slot formed between the body and the seat.

13. The method of claim 12, wherein the slot is formed as a recess at an outer radial surface of the seat assembly.

14. The method of claim 13, wherein the recess extends from an axial end of the seat.

15. The method of claim 9, wherein the seat, the lantern ring, and the seal are positioned in such a way that the seat assembly creates a double piston effect.

16. The method of claim 9, wherein the lantern ring limits movement of the seal such that the seal is prevented from movement to the second end of the seat assembly.

17. A ball valve system, comprising: a body having an upstream end and a downstream end; a ball within the body between the upstream end and the downstream end, the ball being rotatable about a vertical axis to move between an open position and a closed position; a stem with a body and an actuator portion, wherein the body of the stem is positioned substantially parallel to the vertical axis, and wherein the stem is coupled to the ball such that the actuator portion of the stem is configured to rotate the ball between the open position and the closed position; and a seat assembly with a first end facing towards the ball and a second end facing away from the ball, the seat assembly including: a seat, the seat being in contact with the ball; a lantern ring, the lantern ring positioned closer to the second end of the seat assembly than the first end of the seat assembly; and

a seal, the seal positioned between the seat and the lantern ring, wherein the seat, lantern ring, and seal are positioned in such a way that the seat assembly creates a double piston effect, and wherein the lantern ring limits movement of the seal such that the seal is prevented from movement to the second end of the seat assembly.

18. The ball valve system of claim 17, wherein the lantern ring is secured in a slot formed between the body and the seat.

19. The ball valve system of claim 18, wherein the slot is formed as a recess at an outer radial surface of the seat assembly.

20. The ball valve system of claim 19, wherein the recess extends from an axial end of the seat.
